35 — NAÏVE BAYES TEXT CLASSIFIER — SPAM DETECTION

Aim

To implement a Naïve Bayes classifier for classifying SMS messages as spam or ham using text features.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- Pandas
- NumPy
- scikit-learn

Algorithm + Procedure

1. Create a set of SMS messages and their labels.
2. Divide the messages into training and testing sets.
3. Convert text into numerical features using CountVectorizer.
4. Create a MultinomialNB classifier.
5. Train the classifier using the training data.
6. Predict the labels of test messages.
7. Calculate classification accuracy.
8. Display the accuracy.

Code

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import CountVectorizer

from sklearn.naive_bayes import MultinomialNB

from sklearn.metrics import accuracy_score

data = {

    'label': [

        'ham', 'ham', 'spam', 'ham', 'spam',

        'ham', 'spam', 'ham', 'spam', 'ham',
```

```
        'ham', 'spam', 'ham', 'spam', 'ham',
        'spam', 'ham', 'ham', 'spam', 'ham'
    ],
    'message': [
        'Hey are we meeting today',
        'Please call me when you arrive',
        'Congratulations you won a free prize',
        'Can you send me the notes',
        'Win cash now by clicking this link',
        'Are you coming to college today',
        'You have won a free lottery ticket',
        'Please bring your book tomorrow',
        'Claim your free reward now',
        'What time is the class',
        'Lets meet at the library',
        'Congratulations you won a free gift',
        'Please call me later',
        'Free entry in a lucky draw',
        'I will reach home soon',
        'You have been selected for a cash prize',
        'Can you help me with the assignment',
        'The meeting is scheduled tomorrow',
        'Claim your free money now',
        'Please send the project file'
    ]
}
df = pd.DataFrame(data)
X_train, X_test, y_train, y_test = train_test_split(
    df['message'],
```

```
    df['label'],
    test_size=0.3,
    random_state=1,
    stratify=df['label']
)
vectorizer = CountVectorizer()
X_train_v = vectorizer.fit_transform(X_train)
X_test_v = vectorizer.transform(X_test)
model = MultinomialNB()
model.fit(X_train_v, y_train)
pred = model.predict(X_test_v)
print(
    "Accuracy:",
    accuracy_score(y_test, pred)
)
```

Output:

```
Accuracy: 1.0
```